

Weekly Progress Report
Amity School of Engineering & Technology
B. Tech: Program

WPR No: 7	Duration: 31 August 2026 – 6 September 2026
------------------	--

Weekly Progress Report (WPR) for Minor Project

To be filled by Students	
Students Name	Rohit Yadav, Vishal, Hardik Mehlawat (Group 148)
Roll no.	472 (Rohit), 568 (Vishal), 449 (Hardik)
Enrollment no.	A2305223472 (Rohit), A2305223568 (Vishal), A2305223449 (Hardik)
Project Title finalized, if Yes, give name, if NO, give reason	Yes — Deep Learning for AI-Generated Image Detection: A Review of CNN, Vision Transformer, and Explainable Approaches
Synopsis submitted	Submitted (on Amizone, with offer letter)
Literature review	Extraction complete for all 48 primary studies; taxonomy of detection approaches finalised; all 74 reference identifiers verified against arXiv/Crossref records and BibTeX file built
Technical & Economical Feasibility	Not started
Bill of Material	N/A (literature review project this semester)
Project Progress Schedule (PERT Chart)	Week-wise schedule prepared; PERT chart pending
Design of critical components	Paper structure fixed (8 sections); Section 3 taxonomy finalised across subsections 3.1–3.5
Fabrication work (give %)	N/A
Experimental work (give %)	Not started (planned for Semester 8)
Result and Analysis	Not started
Report writing	In progress — Sections 1–3 drafted (about 5,290 words); Section 4 (datasets) and Section 5 (metrics) next
Signature of students	
Work done in this week	
Sections 3.3, 3.4 and 3.5 of the review paper were drafted this week, which completes Section 3. Section 3.3, on transformer and attention-based approaches, runs to 581 words and covers how a vision transformer's patch-based processing differs from convolutional processing when the evidence being looked for is forensic rather than semantic, the one study in the surveyed set that compares a CNN and a ViT under matched conditions — the CNN is better in-domain at 81.1% but falls to 47–57% on unseen manipulation methods, while the ViT holds steadier at about 62% — how pretrained transformers are adapted for the detection task, and the caveat that backbone comparisons in this literature are usually confounded with differences in pretraining and training objective. Section 3.4, on multimodal and foundation-model-based approaches, runs to 617 words and starts from the finding that frozen CLIP features already separate real images from generated ones, then works through the strategies built on that finding — linear probes and few-shot SVMs, intermediate-layer readouts, adapters and prompt tuning, text-side prompts, and subspace decomposition — along with the robustness gaps each of them reports. Section 3.5, on explainability in detection, runs to 616 words and covers Grad-CAM and Grad-CAM++ as the standard tools in this literature, the CIFAKE finding that its classifier keys on background imperfections rather than on the main subject, patch-level detectability maps, prototype-based interpretable classification, and a quantitative framework that compares five explanation methods. Section 3 now totals 3,326 words, and 63 verified references are cited across the draft so far. Every accuracy and dataset figure quoted in the three subsections was copied from the extraction table rather than restated from memory, and two studies whose tests do not amount to cross-generator evidence are described accordingly rather than being counted as such. With Sections 1 (939 words), 2 (1,024 words) and 3 (3,326 words) complete, the draft now stands at roughly 5,290 words. Section 4 (datasets and benchmarks) with the dataset comparison table, and Section 5 (evaluation metrics) with the metrics overview figure, are scheduled next.	
To be filled by Guide (strike off whichever is not applicable)	
Performance of students is satisfactory [☐]	
Performance of students is unsatisfactory [☐]	
A warning to be issued to student(s) (Name) _____	
Student was not well (Name) _____	
Date: 9 September 2026	Signature of Guide

Date: 9 September 2026